Yanbo Cheng

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MASc student in Systems Design Engineering interested in predictive planning, generative models and reinforcement learning for autonomous systems. Projects span RGB-only UAV navigation, camera–LiDAR perception and dynamical-system discovery, with experience in model evaluation, feature ablations and failure analysis.

Education

University of Waterloo Sep 2026–Present
Master of Applied Science, Systems Design Engineering

University of Toronto Aug 2021–Jun 2025
Honours Computer Science Specialist; Mathematics Major

Research and Selected Projects

UAV RGB-Only Navigation Learning 2026–Present

- Implemented RGB-based depth estimation, mapping, DAgger/PPO training, checkpoint reloads and simulator-flight integration.
- Collected simulator flight demonstrations and used flight logs to analyze mapping and goal-grounding behavior during navigation experiments.

Governing-Equation Discovery with Reservoir Computing and SINDy 2024–2026

- Built an echo-state-network augmentation pipeline for Sparse Identification of Nonlinear Dynamics, using Monte Carlo trajectories and uncertainty filtering.
- Evaluated dynamical systems and documented marginal overall gains, negligible benefit for Lorenz, and failure regimes in synthetic-trajectory reliability.

Camera–LiDAR 3D Detection and Feature Painting 2025–2026

- Built a learned difficulty predictor and projected its scores from camera images onto LiDAR points; modified OpenPCDet detection losses and confidence calibration.
- Evaluated PointPillars and SECOND on KITTI validation; compared image, depth and box-size feature ablations for difficulty prediction.

CURI Autonomous Research Runtime 2026–Present

- Built a persistent research runtime that develops questions, executes experiments in isolated worktrees and preserves evidence and decisions across sessions.
- Designed gates against repeated ideas and single-metric hill climbing, with independent reruns of decisive checks before recording outcomes.

Professional Experience

RBC | Software Engineer Intern May–Sep 2023

- Built Power BI/React reporting and Python data pipelines for financial records.

CUSP | Full Stack Developer Sep 2024–2026

- Served as sole developer for a startup's buyer–vendor platform, owning onboarding, role-based dashboards and backend services integrated with Azure and SharePoint.

BalanceAI | Full Stack Developer May–Sep 2024

- Delivered education SaaS features, AI-assisted grading and feedback, and JWT-secured APIs.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL
ML and research: PyTorch, TensorFlow, reinforcement learning, DAgger/PPO, 3D perception, camera–LiDAR fusion, OpenPCDet, scikit-learn, NumPy, Pandas
Engineering: Git, Docker, AWS, Azure, CI/CD, React, Node.js, PySpark